

Spam Email Classification Using Decision Trees and XGBoost with Cross-Validation

ON

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

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by

Name : L. Tejasri roll : 2520090017

Name : Dikshita roll : 2520090007

Under the Esteemed guidance of

Dr. Y. Lakshmi Prasanna

Assistant Professor , CSIT

CSIT



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043

Phone No: 7815926816, www.klh.edu.in





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DEPARTMENT OF _____

Name of Title: Spam Email Classification Using Decision Trees and XGBoost with Cross-Validation

I. Abstract:

Email has become one of the most widely used communication platforms for personal, educational, and business purposes. However, the rapid growth of email usage has also led to an increase in spam messages, which often contain advertisements, phishing links, malware, or fraudulent content. Traditional rule-based spam filters struggle to identify evolving spam patterns, making machine learning a more effective solution for automated spam detection.

This project presents a **Spam Email Classification System** using **Decision Trees** and **Extreme Gradient Boosting (XGBoost)** with **Cross-Validation** to accurately distinguish between spam and legitimate (ham) emails. The system begins with collecting a labeled email dataset, followed by data preprocessing techniques such as text cleaning, tokenization, stop-word removal, and feature extraction using the **Term Frequency-Inverse Document Frequency (TF-IDF)** method. The processed data is then used to train two classification models: Decision Tree and XGBoost.

To ensure reliable model evaluation and reduce the risk of overfitting, **k-fold cross-validation** is employed during the training process. The performance of both models is assessed using standard evaluation metrics, including **accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC score**. A comparative analysis is conducted to determine the strengths and weaknesses of each algorithm, with XGBoost expected to achieve higher classification accuracy due to its ensemble learning approach.

The proposed system provides an efficient and scalable solution for email spam detection while minimizing false positives and false negatives. It demonstrates how advanced machine learning techniques can enhance cybersecurity by protecting users from unwanted and potentially harmful emails. Furthermore, the project can be extended by integrating deep learning models, real-time email filtering, multilingual spam detection, and explainable AI techniques to improve transparency and adaptability. This research highlights the practical application of machine learning in intelligent email filtering systems and contributes to the development of secure and reliable digital communication.

2. INTRODUCTION:

Electronic mail (email) has become one of the most essential communication tools in today's digital world, supporting personal, educational, and business interactions. Millions of emails are exchanged every day, making email services an attractive target for spammers and cybercriminals. Spam emails are unsolicited messages that often contain advertisements, phishing attempts, malware, fraudulent schemes, or malicious links intended to deceive users. These emails not only consume network resources and storage space but also pose significant security and privacy risks.

Traditional spam filtering techniques rely on manually defined rules, blacklists, and keyword-based detection methods. While these approaches are effective against known spam patterns, they often fail to identify newly emerging spam techniques. As cyber threats continue to evolve, there is a growing need for intelligent systems capable of learning from data and adapting to changing spam characteristics. Machine learning offers a powerful solution by automatically identifying patterns in email content and accurately classifying messages as spam or legitimate (ham).

This project proposes a **Spam Email Classification System** using **Decision Trees** and **Extreme Gradient Boosting (XGBoost)** with **Cross-Validation**. Decision Trees provide a simple and interpretable approach to classification by creating decision rules based on input features, whereas XGBoost is a powerful ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting. The use of **k-fold cross-validation** ensures that the trained models are evaluated on multiple subsets of the dataset, resulting in more reliable and unbiased performance estimates.

The project follows a systematic workflow that includes dataset collection, data preprocessing, text cleaning, tokenization, stop-word removal, feature extraction using **Term Frequency-Inverse Document Frequency (TF-IDF)**, model training, testing, and performance evaluation. The models are compared using evaluation metrics such as **accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC score** to identify the most effective spam detection approach.

The primary objective of this project is to develop an accurate, reliable, and scalable spam email classification system that can effectively distinguish between spam and legitimate emails. By leveraging advanced machine learning algorithms, the proposed system aims to improve email security, reduce the impact of unwanted messages, and contribute to safer digital communication. Furthermore, the project provides valuable insights into the practical application of machine learning in cybersecurity and demonstrates the effectiveness of ensemble learning techniques for real-world classification problems.

3. SOFTWARE REQUIREMENTS:

The development of the **Spam Email Classification Using Decision Trees and XGBoost with Cross-Validation** project requires a software environment that supports machine learning model development, data preprocessing, visualization, and model evaluation. The following software tools and libraries are used throughout the project.

3.1 Operating System

- Windows 10/11, Linux (Ubuntu), or macOS

3.2 Programming Language

- Python 3.10 or later

3.3 Development Environment

- Jupyter Notebook
- Visual Studio Code (VS Code)
- Google Colab (optional)

3.4 Python Libraries

- **NumPy** – Numerical computations and array operations.
- **Pandas** – Data loading, cleaning, and manipulation.
- **Scikit-learn** – Machine learning algorithms, TF-IDF vectorization, preprocessing, cross-validation, and evaluation metrics.
- **XGBoost** – Gradient boosting framework for building high-performance classification models.
- **Matplotlib** – Data visualization, including performance graphs and charts.
- **Plotly** – Interactive visualizations for data exploration and result analysis.
- **NLTK (Natural Language Toolkit)** – Text preprocessing, including tokenization, stop-word removal, and stemming.
- **Joblib** – Saving and loading trained machine learning models.

3.5 Dataset

- Spam Email Dataset (such as the SpamAssassin Public Corpus or Kaggle Spam Email Dataset)
- Dataset format: CSV or TXT

3.6 Machine Learning Techniques

- TF-IDF (Term Frequency–Inverse Document Frequency) for feature extraction
- Decision Tree Classifier
- XGBoost Classifier
- K-Fold Cross-Validation

3.7 Model Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix
- ROC-AUC Score

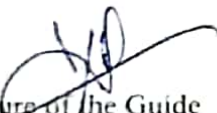
3.8 Optional Deployment Tools

- Streamlit (for developing a simple web interface)
- Flask (for deploying the trained model as a web application)

3.9 Version Control

- Git
- GitHub

The selected software tools provide a reliable and efficient environment for preprocessing email data, training machine learning models, evaluating their performance, and deploying the spam email classification system. These technologies ensure scalability, reproducibility, and ease of future enhancements.



Signature of the Guide



Course Instructor